Drug reaction with eosinophilia and systemic symptoms: Pediatric case series and literature review.

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Pediatric Drug reaction with eosinophilia and systemic symptoms (DRESS) is an uncommon disease that can be difficult to diagnose. This case series and literature review highlights the clinical features of pediatric DRESS and underscores the differential diagnoses, culprit medications, and need for clinical follow-up to detect associated autoimmune sequelae. To describe the clinical and laboratory features of pediatric DRESS, identify associated culprit medications, and discuss the natural history of disease. Ten cases of pediatric DRESS were identified in the electronic medical record by searching the inpatient dermatology consultation list at Indiana University between 2013 and 2018. Clinical and laboratory data were collected including demographics, differential diagnoses, culprit medications, resolution of disease, and autoimmune sequelae. Pediatric patients with DRESS presented at a mean age of 11.5 years and demonstrated a mean time from drug initiation to onset of symptoms of 4 weeks. The most common inciting drugs included antibiotics (62.5%) followed by antiepileptics (37.5%). Rash and transaminitis resolved by 3 weeks, and 20% of patients, all female, developed autoimmune sequelae including Hashimoto's thyroiditis and an undifferentiated connective tissue disorder and occurred at an average of 14.5 weeks after diagnosis. This was a small retrospective study of an uncommon clinical diagnosis at a single institution. Pediatric DRESS was most commonly caused by antibiotics which are being increasingly recognized in the literature as the predominant culprit medications. The development of autoimmune sequelae is a notable consequence that can present weeks after illness and may preferentially affect female patients.

Drug reaction with eosinophilia and systemic symptoms after ocrelizumab therapy.

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We report here on a young woman with multiple sclerosis, who developed a condition with eosinophilia and swelling of limbs seven weeks after initiation of ocrelizumab treatment. We consider her drug reaction to be compatible with a drug reaction with eosinophilia and systemic symptoms (DRESS), also called drug-induced hypersensitivity syndrome. She was treated with antihistamine and corticosteroid treatments, and recovered fully within three months of symptom onset. Ocrelizumab was not re-initiated. We are not aware of other DRESS-like cases related to ocrelizumab treatment.

[DRESS syndrome: 11 case reports and a literature review].

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Study the epidemiological, clinical, biological and chronological drug rash with eosinophilia and systemic symptoms (DRESS) characteristic and indicate the implicated drugs. We carried a retrospective study including all DRESS cases notified to the Pharmacovigilance Unit of Monastir. Our cohort of eleven patients had a median age of 40 years. Clinical examination revealed skin eruption and fever among all patients. Laboratory findings showed marked eosinophilia among all patients, hepatic cytolysis among eight patients and creatinin serum level increase among four patients. An interstitial pulmonary syndrome was noted among two patients. After culprit-drug withdrawal, outcomes were favorable for all

patients. Skin tests were positive with carbamazepin and cefotaxim and negative with sulfasalazine, allopurinol and terbinafine. Throughout this paper, we point out the contribution of skin tests to identify implicated drug in inducing DRESS and to testify cross reactivity and we point out the possibility of neosensitisation to a non related chemical drug after DRESS syndrome.

Drug rash with eosinophilia and systemic symptoms after chlorambucil treatment in chronic lymphocytic leukaemia.

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Drug rash with eosinophilia and systemic symptoms (DRESS) is a severe cutaneous drug reaction with a long duration of eruption and serious organ involvement. The mortality rate has been estimated at about 10%. Aromatic anticonvulsants, sulphamides, minocycline and more rarely carbamazepine are the principal responsible drugs. We report the first case of chlorambucil-induced DRESS syndrome in a 70-year-old man recently diagnosed with chronic lymphocytic leukaemia. He developed recurrent skin rash, fever, hypereosinophilia, and acute renal failure after rechallenge with chlorambucil. The condition improved slowly after stopping medication and systemic steroids. Prompt recognition of a chlorambucil drug reaction is essential in patients receiving chemotherapy.

Drug rash with eosinophilia and systemic symptoms syndrome associated with use of phenytoin, divalproex sodium, and phenobarbital.

PMID: 23456403 | Indexed in PubMed

A probable case of drug reaction with eosinophilia and systemic symptoms (DRESS) associated with consecutive use of three medications for seizure control is reported. A 36-year-old woman was treated at a community hospital for a mild fever (37.9°C) and diffuse raised maculopapular rash with erythema. Three weeks previously, she had been diagnosed with a seizure disorder and initiated on phenytoin (dose unknown) at that time; about two weeks later, she developed a rash, prompting a switch from phenytoin to extended-release divalproex sodium 250 mg orally twice daily. During the week after discontinuation of phenytoin, the rash was improving, but about five days after the initiation of divalproex therapy, she had worsening rash and pruritus requiring urgent treatment; the divalproex was discontinued, and phenobarbital 30 mg three times daily was initiated for continued seizure control. Despite the discontinuation of phenytoin and divalproex, the patient's hepatic function worsened over five days, and phenobarbital therapy was discontinued. With continued deterioration of the patient's condition to fulminant hepatic failure, a transfer to a liver transplant facility was arranged. The use of the adverse reaction probability scale of Naranjo et al. in this case yielded a score of 8, indicating a probable relationship between DRESS and the serial use of phenytoin, divalproex, and phenobarbital. After receiving phenytoin for treatment of seizure disorder, a 36-year-old woman developed a fever and maculopapular rash with erythema. This reaction continued even after drug therapy was switched to extended-release divalproex and then phenobarbital. The patient's liver function deteriorated despite discontinuation of all seizure medications.

Drug reaction with eosinophilia and systemic symptoms/drug-induced hypersensitivity syndrome: clinical features of 27 patients.

PMID: 26271788 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) [also called drug-induced hypersensitivity syndrome (DIHS)] includes severe reactions to drugs that need to be promptly recognized by physicians. To explore heterogeneity in the clinical presentation of DRESS/DIHS at a large academic hospital in Latin America, using the criteria defined by the European Registry of Severe Cutaneous Adverse Reactions (RegiSCAR) scoring system. A retrospective medical record review of 60 patients with diagnostic suspicion of DRESS/DIHS admitted to our hospital between July 2008 and April 2012 was performed, including demographic data, clinical features, laboratory findings and treatment. Of the 60 patients, 27 fulfilled the criteria for DRESS/DIHS. Maculopapular exanthema (85.1%), fever (96.2%) and hepatic involvement (85.1%) were the most common features. Anticonvulsants were the most common causal drugs (77.7%); Phenytoin was the most common individual drug (44.4%), followed by carbamazepine (29.6%). All patients were treated initially with prednisone 1 mg/kg/day. Mortality rate was 4%. The major findings of this study (to our knowledge the largest collection of data on DRESS/DIHS in Latin America) include a positive statistical association between presence of atypical lymphocytes and higher levels of alanine aminotransferase ($P < 0.001$) and reinforce the importance of anticonvulsants in the pathogenesis of this severe reaction.

Management and treatment outcome of DRESS patients in Europe: An international multicentre retrospective study of 141 cases.

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Drug reaction with eosinophilia and systemic symptoms (DRESS) is a rare but potentially fatal drug hypersensitivity reaction. To explore treatment approaches across Europe and their impact on the disease course, as well as prognostic factors and culprit drugs. In this retrospective European multicentric study, we included patients with probable or certain DRESS (RegiSCAR score ≥ 4) between January 2016 and December 2020. Independent associations between clinical parameters and the risk of intensive care unit admission and mortality at three months were assessed using a multivariable-adjusted logistic regression model. A total of 141 patients from 8 tertiary centres were included. Morbilliform exanthem was the most frequent cutaneous manifestation (78.0%). The mean affected body surface area (BSA) was 67%, 42% of the patients presented with erythroderma, and 24.8% had mucosal involvement. Based on systemic involvement, 31.9% of the patients had a severe DRESS. Anticonvulsants (24.1%) and sulphonamides (22.0%) were the most frequent causative agents. In all, 73% of the patients were treated with systemic glucocorticoids, and 25.5% received topical corticosteroids as monotherapy. Few patients received antiviral drugs or anti-IL5. No patients received intravenous immunoglobulins. The overall mortality was 7.1%. Independent predictors of mortality were older age (≥57.0 years; fully adjusted OR, 9.80; 95% CI, 1.20-79.93; $p = 0.033$), kidney involvement (fully adjusted OR, 4.70; 95% CI, 1.00-24.12; $p = 0.049$), and admission in intensive care unit (fully adjusted OR, 8.12; 95% CI, 1.90-34.67; $p = 0.005$). Relapse of DRESS and delayed autoimmune sequelae occurred in 8.5% and 12.1% of patients, respectively. This study underlines the need for diagnostic and prognostic scores/markers as well as for prospective clinical trials of drugs with the potential to reduce mortality and complications of DRESS.

Drug reaction with eosinophilia and systemic symptoms (DRESS) in the pediatric population: A systematic review of the literature.

PMID: 32247873 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) is a drug-induced hypersensitivity reaction that can have fatal complications. Although substantial data exist regarding DRESS in adults, to our knowledge, a systematic review of available literature has not been performed in children. To review available data on DRESS in the pediatric population. A systematic literature review was performed for pediatric (aged <18 years) patients with DRESS. We included 82 articles with 148 patients; of these, 97.9% experienced a skin rash, and the liver was the second most common organ involved (84.5%). Among 143 patients for which a treatment regimen was reported, 85.3% were treated with systemic steroids. Intravenous immunoglobulin alone failed to improve symptoms in 5 patients who were initially misdiagnosed, whereas those treated with intravenous immunoglobulin and steroids (2.7%) showed rapid clinical improvement. The mortality rate was low (3.0%). Complications included multiorgan failure and acute respiratory distress syndrome. Limitations included limited availability of data for statistical analysis. Pediatric DRESS commonly involves the liver. With treatment, the prognosis is commonly good, but serious complications may occur. Corticosteroids, possibly in conjunction with intravenous immunoglobulin in severe cases, may serve as an effective, valuable treatment of pediatric DRESS.

Histopathological analysis and clinical correlation of drug reaction with eosinophilia and systemic symptoms (DRESS).

PMID: 24329105 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) is a severe cutaneous adverse drug reaction. However, its histopathological features have not been well defined. To identify the clinicohistopathological findings of DRESS, and analyse the cutaneous histopathological changes observed in DRESS compared with those observed in maculopapular exanthema (MPE). In a retrospective study, conducted at Chang Gung Memorial Hospital (Taiwan) between 2001 and 2011, we compared the clinicohistopathological features of 32 patients with probable/definite DRESS (defined by the RegiSCAR scoring system) with those of 17 patients with MPE. The major pathological changes observed in patients with DRESS included dyskeratosis (97%), epidermal spongiosis (78%), interface vacuolization (91%), perivascular lymphocytic infiltration (97%) and eosinophilic infiltration (72%). Many pathological features were common to both MPE and DRESS. However, severe dyskeratosis, epidermal spongiosis and severe interface vacuolization were significantly more prominent in cases of DRESS ($P < 0{\cdot}05$). The presence of severe dyskeratosis was significantly associated with the clinical severity of renal impairment ($P = 0{\cdot}01$). The severe dyskeratosis detected in patients with DRESS may correlate with a greater extent of systemic involvement compared with that noted in MPE. However, the histopathological changes associated with DRESS are not entirely specific.

Sildenafil-induced drug reaction with eosinophilia and systemic symptoms.

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Drug reaction with eosinophilia and systemic symptoms (DRESS)/drug-induced hypersensitivity syndrome

(DIHS) is a life-threatening, multi-organ adverse drug reaction with a mortality rate of approximately 10 %-20 %. The most common culprit drugs are anticonvulsants, some antibiotics such as dapsone and minocycline, salazosulfapyridine, allopurinol and some antiretroviral molecules such as abacavir and nevirapine. Only one case of DRESS induced by sildenafil has been reported in the literature. Here we report a new case.